

Application of Approximation Theory in Data Science

Viewing Machine Learning through the Lens of Numerical Analysis

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Introduction: Connecting Math to Reality

In quantitative analysis, we face a constant challenge: Reality is complex, but we need simple functions to model it. This is exactly what **Approximation Theory** is about.

While it has roots in Numerical Analysis, for us in Data Science, it boils down to two main tasks:

1. **Simplification:** Turning heavy functions (like complex exponentials) into easy-to-handle polynomials.
2. **Data Fitting:** Finding a curve that best represents a set of scattered data points.

In modern Data Science, “Data Fitting” isn’t just theory—it is the backbone of **Supervised Learning**.

The Discrete Problem: Linear Algebra and Regression

Let’s start with a classic scenario: we have a dataset comparing **Marketing Budget** (x) to **Sales** (y). We want to find a function $f(x)$ so that for our data, $f(x_i) \approx y_i$.

If we assume the relationship is linear, we are essentially trying to solve this system:

$$\begin{cases} ax_1 + b = y_1 \\ \vdots \\ ax_m + b = y_m \end{cases} \implies A\mathbf{x} = \mathbf{b}$$

The Overdetermined System

Here is the catch: in data science, we almost always have more data points (m) than variables (n). This creates an **overdetermined system**, meaning there is no exact solution for $Ax = b$.

To fix this, we use **Discrete Least Squares**. Instead of chasing an impossible exact solution, we minimize the error (the Euclidean norm of the residual $\mathbf{r} = \mathbf{b} - A\mathbf{x}$):

$$\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2$$

Calculus gives us the answer where the gradient is zero, leading to the famous **Normal Equations**:

$$A^T A \mathbf{x} = A^T \mathbf{b}$$

This single equation is the engine behind Linear Regression and many machine learning techniques we use daily.

Feature Engineering and the Vandermonde Matrix

Often, a straight line doesn't cut it. We might suspect a quadratic relationship: $P(x) = a_0 + a_1x + a_2x^2$. This allows us to ask deeper questions:

1. What are the base sales? (a_0)
2. How does the budget drive sales? (a_1)
3. Does the budget lose effectiveness over time? (a_2)

To solve this, we build the **Vandermonde Matrix** to hold our features:

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_m & x_m^2 \end{bmatrix}$$

Interpretation: If a_2 turns out negative, we have mathematically proven “Diminishing Returns.”

Visualizing the Bias-Variance Tradeoff

Choosing the right model complexity is tricky.

1. Underfitting (High Bias): If the model is too simple (Degree 1), we miss the pattern entirely.

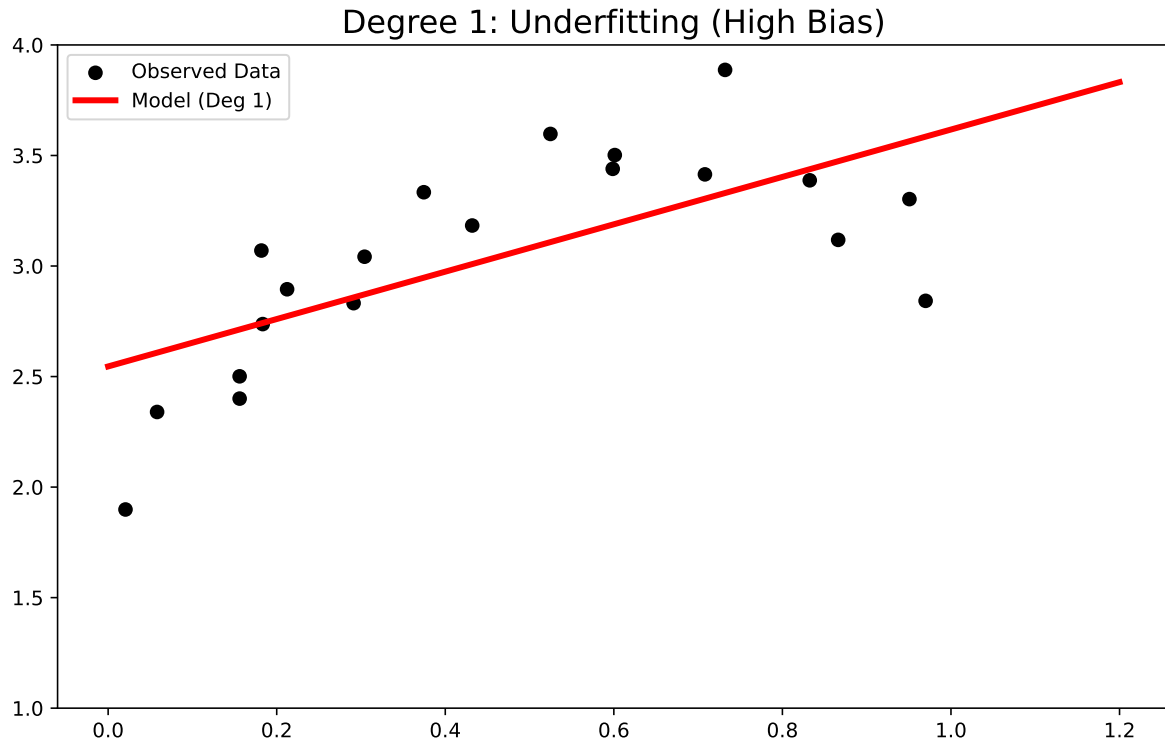


Figure 1

2. Optimal Fit: A degree 2 polynomial hits the sweet spot, capturing the trend without memorizing noise.

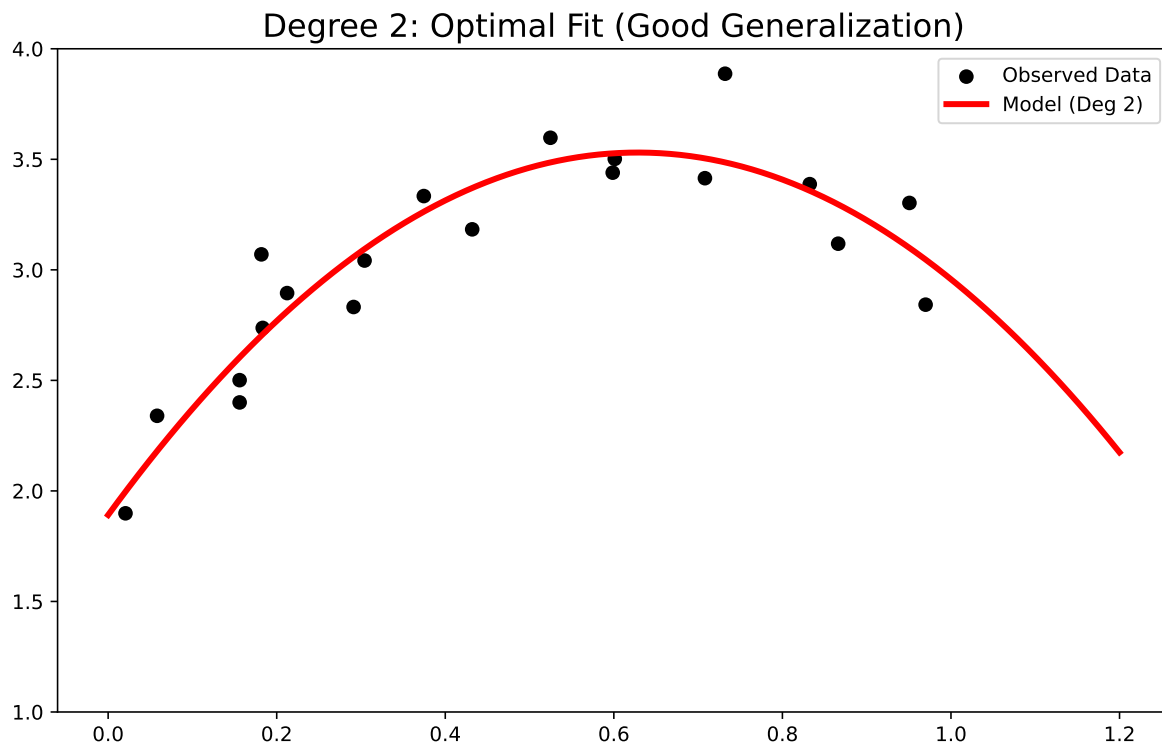


Figure 2

3. Overfitting (High Variance): If the degree is too high (e.g., 15), the model connects the dots blindly, capturing noise instead of the signal.

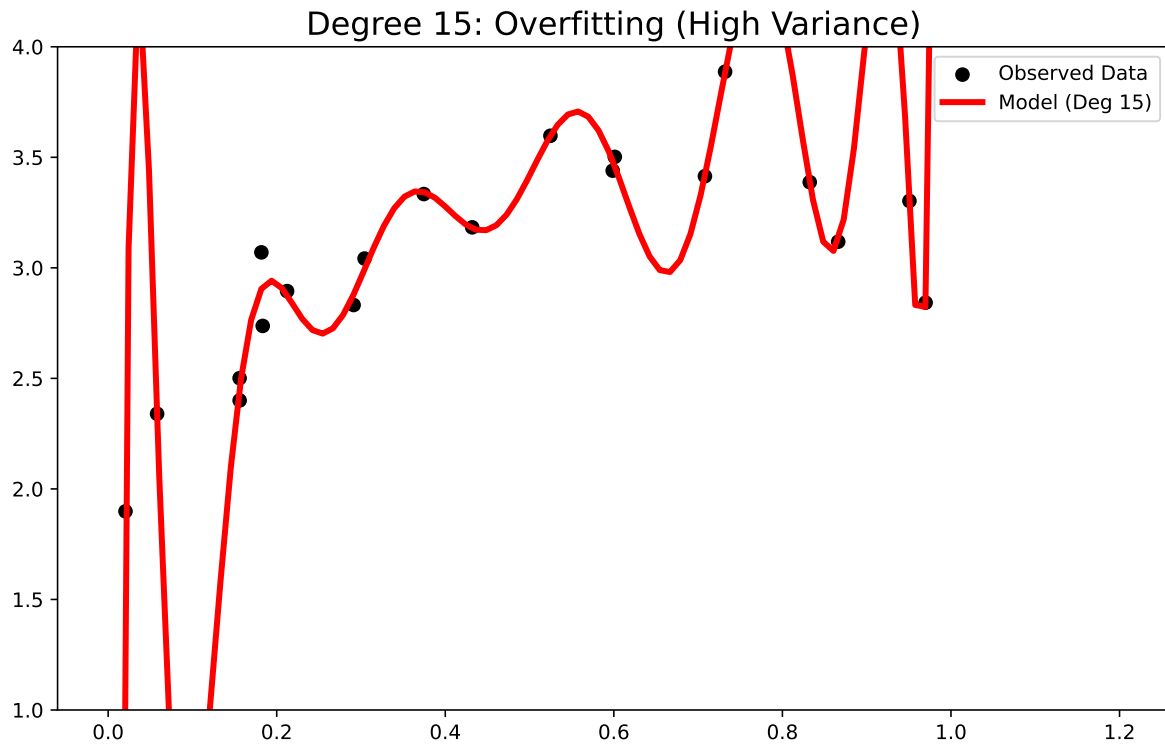


Figure 3

Selecting the Best Model

To find the right degree (n), we look at the **Mean Squared Error (MSE)**. As complexity goes up, training error always drops, but test error (generalization) will eventually spike.

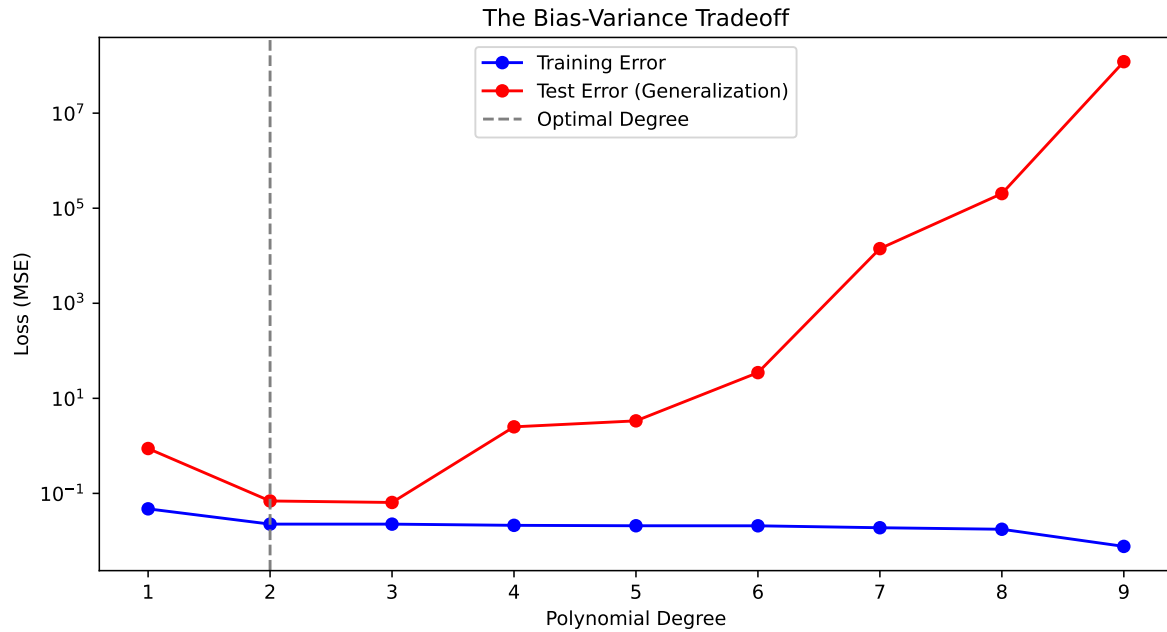


Figure 4

Beyond 2D: Multiple Regression

Real-world problems rarely have just one variable. Suppose we add **Competitor's Price** (z). Now, our model is a plane in 3D: $y = a_0 + a_1x + a_2z$.

Mathematically, nothing changes. We solve the same normal equations, just with a larger matrix A .

Fitting a Plane (Multiple Regression)

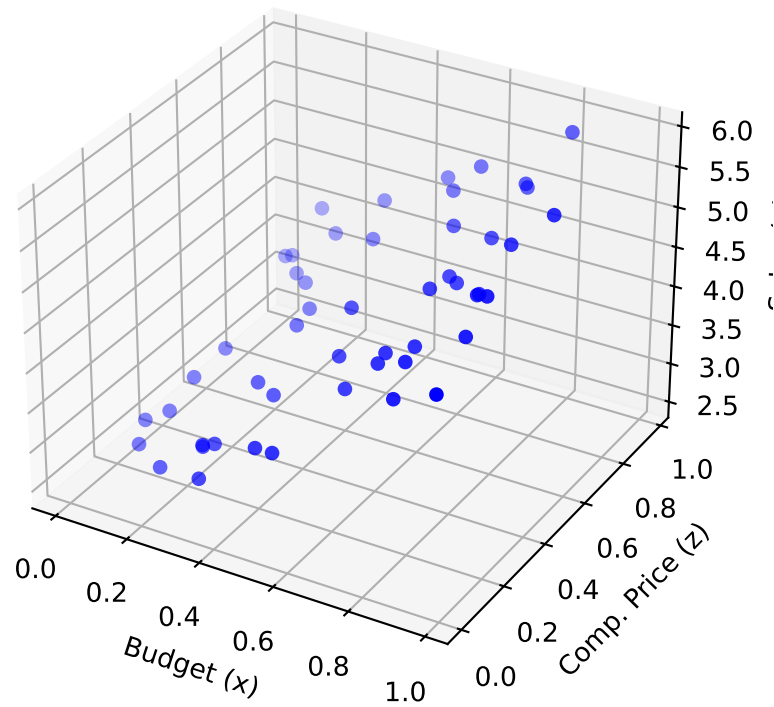


Figure 5

The Continuous Problem: Signal Processing

Discrete data is standard in business, but fields like audio processing or ECG analysis deal with **continuous signals**. Here, we want to approximate a complex function $f(x)$ with a polynomial for compression or speed.

Minimizing the Integral

Since we are dealing with functions, not points, we switch from summation to integration. We minimize the **Integral of Squared Error**:

$$E = \|f - P_n\|_2^2 = \int_a^b (f(x) - P_n(x))^2 dx$$

The Trap: Standard Basis vs. Orthogonality

A beginner might try the standard basis $(1, x, x^2, x^3)$. Here is the problem: on the interval $[0, 1]$, x^2 and x^3 look very similar. They are highly correlated (multicollinearity). This confuses the model and makes coefficients unstable.

The fix is **Orthogonality**. We need features that are independent, meaning their inner product is zero:

$$\langle \phi_i, \phi_j \rangle = \int_a^b \phi_i(x) \phi_j(x) dx = 0 \quad (i \neq j)$$

Orthogonal bases (like Legendre Polynomials) give us:

1. **Decoupling:** We can calculate each coefficient separately.
2. **Stability:** Adding a higher-degree term doesn't mess up the previous ones.

Final Simulation: Reconstructing a Complex Signal

To show the power of Continuous Least Squares, let's reconstruct a complex signal (mixed sine, cosine, and exponentials) using Legendre polynomials. Because of orthogonality, we can compute the "contribution" of each degree independently.

This process highlights the elegance of the math: by projecting complex realities onto simpler orthogonal structures, we get both speed and clarity.

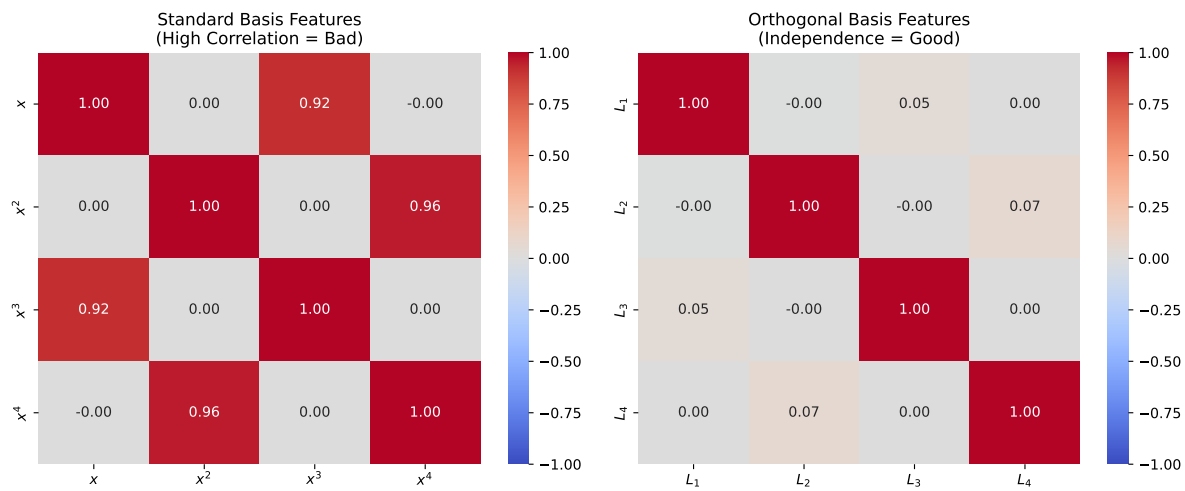


Figure 6

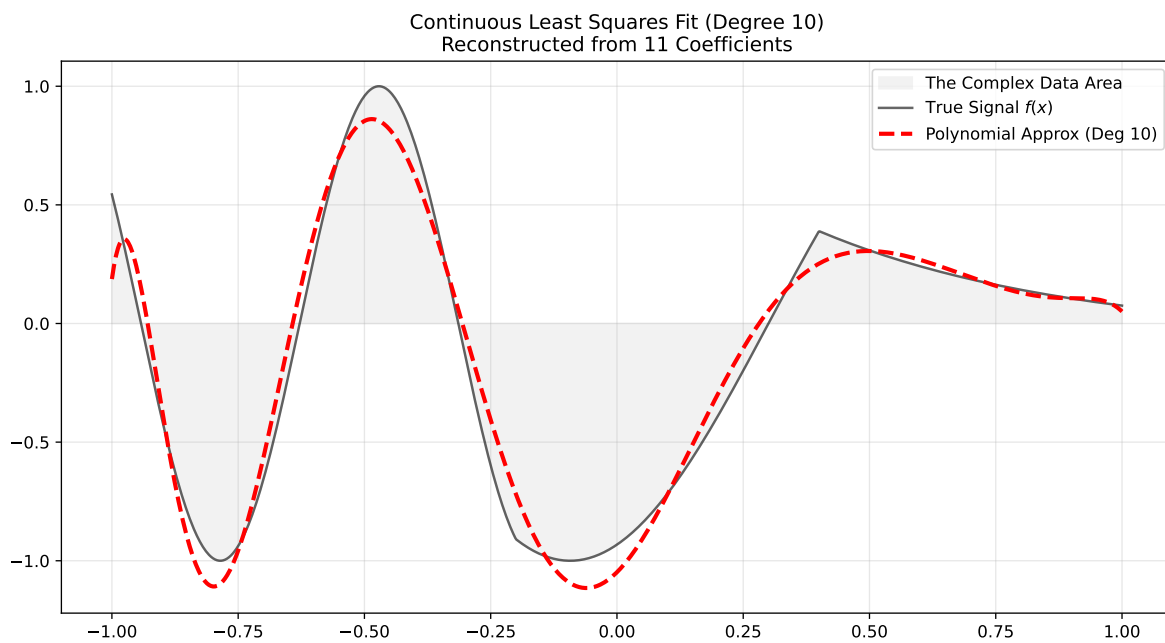


Figure 7